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MILITARY SPECIFICATION

TORCHES, OXYACETYLENE, CUTTING AND WELDING

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers hand-held, oxyacetylene torches (assemblies of torch handles, cutting attachments, welding and cutting tips, or single unit cutting torches) used for welding and cutting ferrous metals.

1.2 Classification. The torches shall be of the following types, as specified (see 6.2.1):

Type I	Torch, combination, welding, with cutting attachment. (To weld metal thicknesses 1/32 thru 1/2 inch and cut metal thicknesses 1/8 thru 6 inches)
Type II	- Torch, cutting. (To cut metal thicknesses 1/8 thru 12 inches)

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications and standards. The following specifications and standards form a part of this specification to t-he extent specified herein. Unless otherwise specified, the issues of these documents shall be those listed in the issue of the Department of Defense Index of Specifications and Standards (DoDISS), and supplement thereto, cited in the solicitation.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, US Army Armament Research and Development Center, ATTN: SMCAR-EST-S (R), Rock Island, IL 61299-7300 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

FSC 3433

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SPECIFICATIONS

MILITARY

MIL-C-5501 - Caps and Plugs, Protective, Dust and Moisture Seal, General Specification for.
MIL-W-45562 - Welding and Soldering Equipment, Supplies, and Accessories, Packaging of.

STANDARDS

FEDERAL

FED-STD-H28 - Screw-Thread Standards for Federal Services.

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-130 " Identification Marking of US Military Property.
MIL-STD-889 - Dissimilar Metals.

2.1.2 Other Government documents. The following other Government documents form a part of this specification to the extent specified herein. Unless otherwise specified, the issues shall be those in effect on the date of the solicitation.

US DEPARTMENT OF LABOR

OSHA 2206 - General Industry, OSHA Safety and Health Standards (29 CFR 1910).

(Copies of specifications, standards, and documents required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting activity.)

2.2 Other publications. The following document(s) form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of the documents which are DoD adopted shall be those listed in the issue of the DoDISS specified in the solicitation. Unless otherwise specified, the issues of documents not listed in the DoDISS shall be the issue of the nongovernment documents which is current on the date of the solicitation.

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

A5TM 3572-81 - Standard Test Method for iiubber Deterioration by Heat and Oxygen.
ASTM D3951-82 - Standard Practice for Commercial Packaging.

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

Compressed GAS ASSOCIATION (CGA)

CGA E-1 - Standard Connections for Regulator Outlets, Torches and Fitted Hose for Welding and Cutting Equipment.

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CGA E-2 - Hose Line Check Valve Standards for Welding and Cutting.

(Requests for copies should be addressed to the Compressed Gas Association, Inc., 1235 Jefferson Davis Hwy, Arlington, VA 22202.)

COPPER DEVELOPMENT ASSOCIATION (CDA)

Standards Handbook, Wrought Copper and Copper Alloy Hill Products

STANDARD DATA SHEETS

Part 2 - Alloy Data:
Copper No. 110 - Electrolytic Tough Pitch.
Copper No. 122 - Phosphorous Deoxidized, High Residual Phosphorous.
Copper No. 145 - Phosphorous Deoxidized, Tellurium Bearing.
Copper No. 147 - Sulfur Bearing.

(Requests for copies should be addressed to the Copper Development Association, Inc., Greenwich Office Park 2, BOX 1840, Greenwich, CT 06836.)

UNDERWRITERS LABORATORIES (UL)

UL 123 - Oxy-Fuel Gas Torches.

(Requests for copies should be addressed to the Underwriters Laboratories, 333 Pfingsten Road, North Brook, IL 60062.)

(Nongovernment standards and other publications are normally available from the organizations which prepare or which distribute the documents. These documents also may be available in or through libraries or other informational services.)

2.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein (except for associated detail specifications, specification sheets, or MS standards), the text of this specification shall take precedence. Nothing in this specification, however, shall supersede applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 First article. When specified in the contract or purchase order. a sample shall be subjected to first article inspection (see 4.3 and 6.2.1).

3.2 UL listing. All torches furnished under this specification shall meet the requirements of Underwriters Laboratories Standard UL 123. All welding torches, cutting attachments, and cutting torches submitted under this specification shall be listed by Underwriters Laboratories and shall be labeled to indicate such, in accordance with UL labeling requirements.

3.3 Design. All torches shall be hand-held oxyacetylene torches, of the positive pressure type, incorporating facilities for mixing and controlling the flow of gases to the torch tip (see 6.4).

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3.3.1 Type I. Type X torches shall meet all requirements of Section 3 of this specification, except those requirements specifically identified as pertaining to Type II torches only. Type I torches shall consist of a welding torch handle as specified in 3.5.1 and a cutting attachment as specified in 3.5.5 and, in addition, shall be supplied with welding tips and mixer(s) or welding tip assemblies as specified in 3.5.4 and cutting tips as specified in 3.5.5.5.

303.2 Type 11. Type IX torches shall be single unit cutting torches and shall be supplied with cutting tips as specified in 3.5.5.5. Type 11 cutting torches shall meet all of the requirements of Section 3 of this specification, except that Type II cutting torches shall he an assembly consisting of the torch body , gas tubes, and cutting torch head. Type II torches shall be no less than 18 inches and no more than 24 inches long. No welding tips, welding tip assemblies, or cutting attachments shall be supplied for use with Type 11 torches.

3.3.3 Construction. All parts used in fabrication of the torches shall be new, clean, and free of oil and grease. In no event shall patching processes be used for reclaiming defective parts. Surfaces of all parts shall be free of burrs, defects, and dirt. Parts subject to removal or adjustment shall not be permanently deformed by the assembly process. Torches shall not be assembled from parts reclaimed from used torches.

3.3.3.1 Threads. Screw threads shall comply with the requirements of FED-STD-H28 and the Detailed Standard sections referenced therein. Threads for hose connections shall comply with Compressed Gas Association Standard CGA E-1, connections 022 and 023. All internal and external threads shall be clean, free of any extraneous materials, and shall have no burrs or nicks.

3.3.3.2 Interchangeability. All parts shall be manufactured to definite standards, clearances, and tolerances in order that any such parts of a particular type or model may be replaced, interchanged, and adjusted without modification of the replacement parts or any other parts of the unit. When practical, all parts shall be permanently and legibly marked in accordance with MIL-STD-130.

3.3.3.3 Brazing. Brazing shall be of the highest quality workmanship prevailing among commercial manufacturers normally producing torches of the types specified herein. Flux shall be cleaned from surfaces after brazing is completed. Brazing filler metal shall have a melting temperature no lower than 1000° Fahrenheit (F). Filler metal shall be visibly complete around all brazed joints.

3.3.3,4 Machined surfaces. Gas holes and passages shall be free of burrs. Machined surface seals shall be free of nicks, burrs, and scratches and shall meet the leakage requirements of paragraph 3.4.3.

3.3.3.5 Lubricants. Lubricants inside the torch shall be compatible with oxygen at 90 pounds per square inch gage (psig).

3.3.3.6 Material. Materials not specifically designated herein or in the contract shall be of a quality commensurate with commercial practice within the producing industry, shall be suitable for the intended purpose in the design of the end item, and shall meet all requirements specified herein. Materials shall be free from defects which would adversely affect the performance or

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maintainability of the individual components or the overall assembly. It shall not be permissible to use reclaimed parts as is, or rebuilt from scrap or other used equipment. When dissimilar metals are used in contact with each other, suitable protection against galvanic corrosion shall be applied in accordance with MIL-STD-889.

3.4 Performance and product characteristics.

3.4.1 Capacities. Torch flow rate capabcities shall conform to the requirements specified in 3.4.1.1 and 3.4.1.2.

3.4.1.1 Oxygen flow rate. A Type I torch, with its largest cutting tip (see 3.5.5.5) installed in the cutting attachment, shall have a cutting oxygen flow rate of 450 cubic feet per hour (cfh) or more at a torch oxygen inlet pressure not exceeding 90 psig. A Type II torch, with its largest cutting tip installed, shall have a cutting oxygen flow rate of 850 cfh or more at a torch oxygen inlet pressure not exceeding 85 psig.

3.4.1.2 Acetylene flow rate. Type I torches, with the largest cutting tip installed in the cutting attachment and the preheat flame adjusted to neutral, shall have an acetylene flow capacity of at least 25 cfh at a torch acetylene inlet pressure not exceeding 10 psig. Type II torches, with the largest cutting tip installed and the preheat flame adjusted to neutral, shall have an acetylene flow capacity of at least 40 cfh at a torch acetylene inlet pressure not exceeding 10 psig.

3.4.2 Backfire and flashback. Each Type I and Type 11 torch mix cutting torch and each tip mix cutting tip shall withstand five backfires (see 6.4), at a rate of at least one per second with no flashback (see 6.4), when the oxygen and acetylene are set at the manufacturer’s recommended pressures and the preheat flame is adjusted to neutral. WARNING: A flashback condition necessitates immediately cutting off the gas supply (oxygen first, acetylene second) to prevent damage and possible destruction of the torch, gas tubes, or hoses.

3.4.3 Leakage. Joints, valves, O-ring seals, metal to metal mating seals, gas passages, connections, and glands shall be capable of sustaining gas pressures of at least 50 pounds psig internally (from one gas passage to another) and externally (from internal gas passage to external atmosphere). Valves shall sustain the gas pressures without leakage sufficient for formation of a bubble when submerged in clean water for ten seconds, both when the valves are closed and when the valves are open to gas passage. The cutting tip seats shall not allow gas leakage at operating pressures, or higher, when tightened with no more than 20 foot-pounds of torque applied to the tip nut. The welding tip seats shall not allow gas leakage at operating pressures, or higher. when hand tightened.

3.5 Details of components.

3.5.1 Torch handle. The Type I torch handle shall consist of a grip, gas passages, acetylene inlet valve and hose connection, oxygen inlet valve and hose connection, and front body. The Type I torch handle may also include the cutting oxygen valve in which case the oxygen valve on the torch handle may serve as the preheat oxygen control valve during cutting (see 3.5.5). The Type II torch handle shall be firmly assembled to the rest of the torch in such a manner that it cannot be removed without the use of tools. The Type II torch handle shall

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consist of all the above except for a front body. Non-gas carrying spaces in the handle shall be vented. The torch handle may also be constructed with the gas passages drilled or formed through a solid body handle, or as a tube within a tube.

3.5.1.1 Grip. The grip of the torch handle shall be knurled, grooved, or ribbed for at least four inches of its length. Grips that are part of a handle not constructed with gas passages drilled or formed through a solid body handle shall be fixed to the handle with pins, screws, threads, solder, or brazing metal.

3.5.1.2 Preheat control valves. The oxygen and acetylene control valves shall be permanently and legibly marked on the valve body, the valve control knob, or the torch body or handle at a point adjacent to the inlet in accordance with the instruction for marking gas inlets specified in 3.5.1.2.5. If the valve knobs are marked, the valve stems shall not be interchangeable. Valve stems shall be equipped with packing and shall meet the leakage requirements of 3.4.3. The valve stem packing shall be on the don-stream side of the valve. Each control valve seat shall provide a seal that does not allow leak-through at 50 psig or more, from the up-stream side of the valve to the down-stream side of the valve, when the valve is closed with no more than 20 inch-pounds of torque applied to the knob. Valve seats shall be constructed so that seating occurs between materials of different hardness.

3.5.1.2.1 Valve bodies. The valve bodies shall be forged, cast, or constructed from bar stock. Valve bodies shall be formed separately or as integral parts of the torch handle. Valve bodies which are formed separately shall be attached to the torch handle and shall meet the requirements of 3.4.3.

3.5. 1.2.2 Valve stems. Valve stems shall be fabricated of corrosion-resistant metal. Valve stems shall maintain a set valve adjustment during operation of the torch, and shall not be moveable by accident-ally brushing with sleeve or glove. Stems of the oxygen and acetylene valves shall be assembled to prevent removal from the valve without first removing the gland nut and packing.

3.5. 1.2.3 Valve packing. The valve packing shall not leak during torch operations. The Packing, or alternate means, shall provide a drag equivalent to at least four inch-ounces of torque on the valve stem to aid the stem in maintaining its setting during use, and to prevent the stem from being turned by accident.

3.5. 1.2.4 Valve knobs. Valve knobs shall be fabricated of corrosion-resistant metal. Valve knobs shall be at least 0.875 inch in diameter and shall provide a positive grip for an operator wearing welding gloves. Valve knobs shall not separate from the stem when pulled with a force of at least 70 pounds along the centerline of the stem. Valve knobs shall not , lossen or separate from the stem when at least 50 inch-pounds of torque is applied to the knob in both the clockwise and counter-clockwise directions.

3.5. 1.2.5 Gas inlets. Acetylene and oxygen inlets shall have threaded connections that comply with the requirements of CGA E-1, connections 022 and 023. Each inlet shall be permanently marked to indicate acetylene or fuel and oxygen. The minimum allowed marking shall be no less than an ‘*A” or “F”, and "O". Hose connections shall be at. the rear of the torch handle. All threads shall be

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clean and free of dirt. Conical seats shall be smooth, symmetrical, and free of marks, nicks, and corrosion.

3.5.2 Reverse flow check valves. Two UL listed reverse flow check valves, conforming to CGA E-2, one for each inlet valve, shall be provided with each torch handle. When attached to the torch, the mating fit of the reverse flow check valves shall meet the requirements of 3.4.3. UL listed torches with integral check valves are acceptable.

3.5.3 O-rings. O-rings shall not crack or show visible evidence of deterioration following exposure for 96 hours to oxygen, at a pressure of 300 psig (2070 kpa) and at a temperature of 158°F (70°C), when tested in accordance with ASTM D572-81. O-rings shall not fail to provide adequate leak free seals, in accordance with 3.4.3, due to chemical attacks of the gases used, or because of normal operating and environmental conditions.

3.5.4 Welding tip assembly. Welding tip assemblies shall have single flame tips conforming to 3.5.4.1 and shall consist of a mixing chamber which attaches to the welding torch handle and a welding tip which attaches to, or is integral with, the mixer. All sealing and mating surfaces shall provide a seal that complies with the requirements of 3.4.3. No O-rings shall be used between the mixing chamber and the flame. The welding tip shall have a firm, metal-to-metal joint with the mixer.

3.5.4.1 Welding tips. Unless otherwise specified (see 6.2.1), one each of the following sizes of welding tips shall be supplied with each welding torch. Welding tips shall be fabricated of copper conforming to CDA Standard Data Sheet for Copper No. 122 and shall have a thermal conductivity of not less than 85 percent of pure electrolytic tough pitch copper. Tips shall be legibly and permanently identified for size according to the manufacturer's standard commercial practice. If welding tips are not integral with their mixers, the tips shall be threaded externally to provide a positive, gas tight, metal-to-metal fit to the mixer. Welding tips shall be of the goose neck type. Welding tips shall provide a long or bulbous flame symmetrical with the axis of the tip end. Welding tip orifice sizes shall fall within the ranges specified as follows:

Number Drill Size	Diameter (Inches)
70 - 74	0.028 - 0.0225
55 - 62	0.052 - 0.038
40 - 45	0.098 - 0.082

3.5.5 Cutting attachments. Cutting attachments shall consist of a mating part and retaining nut for mounting to the welding torch handle, gas tubes for carrying gases to the tip, a cutting torch head, a tip retaining nut, a preheat oxygen control valve, and a lever operated valve to control cutting oxygen. The preheat oxygen valve and the cutting oxygen valve are not required in the cutting attachment if those functions are provided in the welding torch handle (see 3.5.1). Type I cutting attachments shall be mountable onto the Type I handle with a gas tight fit, conforming to 3.4.3, by hand tightening only and without the use of a tool. Type II cutting torches shall meet the requirements of 3.S.5.1, 3.5.5.2, 3.5.5.3, and 3.5.5.4.

3.5.5.1 Gas tubes. Gas tubes in the Type I cutting attachment and the Type II cutting torch shall not permanently deform, nor shall the brazed joints fail

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when a wrench is used to tighten the tip nut with no less than 40 foot-pounds of torque applied to the tip nut at 90 ±5 degrees to the tubes, and with the tubes supported no closer than five inches from the centerline of the tip nut. Gas tube joints shall be brazed to the cutting head. Brazing shall conform to 3.3.3.3.

3.5.5.2 Cutting head. The Type I cutting attachment head and the Type II cutting torch head shall be fabricated of copper alloy or stainless steel, and shall have a 90 +2 degree angle of the tip position relative to the centerline of the welding torch handle. The cutting head shall incorporate a tip nut that shall hold the tip against the seat of the cutting head.

3.5.5.3 Cutting oxygen control valve. The cutting oxygen control valve shall not leak externally at 50 psig, or more, when open and when closed. The cutting oxygen control valve shall not leak through from the up-stream side of the valve to the down-stream side of the valve when subjected to 50 psig or more with the valve closed. All of these conditions shall be met after the cutting oxygen control valve has been opened and closed by the lever at least 1000 times. There shall be no bent, broken, or loose parts after at least 1000 cycles of operation.

3.5.5.4 Lever. The cutting oxygen valve lever shall be fabricated of corrosion-resistant metal. Levers shall not permanently deform under 30 pounds or more of force, when applied toward the centerline of the torch handle at least four inches from the lever fulcrum or next nearest supporting point to the end of the lever, or within 1/2 inch of the end of the lever if less than four inches long.

3.5.5.5 Cutting tips. Cutting tips shall be fabricated of copper conforming to CDA Standard Data Sheets for Copper No. 110, 145, or 147 and shall have a thermal conductivity of not less than 90 percent than that of the thermal conductivity of pure electrolytic copper. Cutting tips shall be one piece, designed for use with acetylene, and shall be configured with at least four preheat holes and one cutting oxygen hole. The tips shall be permanently and legibly identified for size according to the manufacturer's standard commercial practice. Cutting tip seats shall provide a tight seal to prevent gas leaks around the tip nut and to prevent cross-over leaks from the cutting oxygen line to the preheat oxygen line or mixed preheat gas line. The seal between the cutting tip and the cutting tip seat shall meet the requirements of 3.4.3. Cutting tip orifice sizes shall fall within each range specified below. The center cutting oxygen orifice shall provide a stream of oxygen which travels down the centerline of the preheat flame group and which shall not deviate from the centerline by more than two degrees. When a MAPP tip(s) is required, the acetylene tip which it is replacing shall be identified (see 6.2.1). Unless otherwise specified (see 6.2.1), one each of the following four sizes of tips shall be supplied with each Type I torch:

Preheat Hole Drill Size Number	Diameter (Inches)	Oxygen Hole Drill Size Number	Diameter (Inches)
62 - 75	.038 - .021	67 - 56	.032 - .0465
58 - 71	.042 - .026	57 - 53	.043 - .0595
56 - 69	.0465 - .0292	51 - 43	.063 - .089
53 - 69	.0595 - .0292	43 - 31	.089 - .120

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Unless otherwise specified (see 6.2.1), one each of the above four sizes of tips and one of the following size tip shall be supplied with each Type II torch:

Preheat Hole Drill Size Number	Diameter (Inches)	Oxygen Hole Drill Size Number	Diameter (Inches)
53 - 69	.0595 - .0292	31 - 20	.120 - .161

3.5.5.6 Mixing. Gas mixing shall occur in the cutting attachment, cutting torch, or the tip.

3.6 Wrench. Each torch shall be supplied with a wrench(s) having suitable openings for tip nuts, hose connections, and packing nuts where adjustment is required. The wrench(s) shall be plated or treated for corrosion resistance or shall be fabricated of material that is inherently corrosion resistant.

3.7 Safety and health requirements The torches shall comply with the general safety requirements promulgated under OSHA 2206 that are applicable to the torches themselves. Exceptions and additional requirements for safety and health shall be as specified in the contract (see 6.2.1 and 6.5).

3.8 Product identification. The torches shall be marked for product identification in accordance with MIL-STD-130 and, unless otherwise specified (see 6.2.1), shall include the National Stock Number. In addition, unless otherwise specified (see 6.2.1), all welding torch handles, cutting attachments, and cutting torches shall be permanently marked with the manufacturer's name and model number.

3.9 Workmanship. Workmanship of the torches and accessories shall be of the highest quality prevailing among manufacturers normally producing torches of the type specified herein for the commercial market.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance or all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items must meet all requirements of Sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of assuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling in quality conformance does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to acceptance of defective material.

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4.2 Classification of inspection. The Inspection requirements specified herein are classified as follows:

(a) First article inspection (see 4.2.1)
(b) Quality conformance inspection (see 4.2.2)
(c) Comparison tests (see 4.2.3)

4.2.1 First article inspection. First article inspection shall be performed on a preproduction sample(s) or first production item(s) (see 3.1). Unless otherwise specified (see 6.2.1), first article inspection shall consist of the examination in 4.3 and the tests of 4.4.1 through 4.4.7. Failure of the first article to pass any examination or test shall be cause for rejection.

4.2.2 Quality conformance inspection.

4.2.2.1 Inspections and tests. Quality conformance inspection shall be applied to production units offered for acceptance under the contract. Quality conformance inspection shall consist of (a) through (d) below. Failure of any unit to pass an examination or test shall be cause for rejection of the unit.

(a) Product examination (see 4.3)
(b) Leak tests (see 4.4.2)
(c) Flame tests (see 4.4.3)
(d) Packaging inspection (see 4.5)

4.2.2.2 Sampling. Sampling for quality conformance inspection shall be performed in accordance with MIL-STD-105. Sampling for product examination shall be at Level II with a minimum sample of 20 and no defects allowed. Sampling for packaging inspection shall be at Special Inspection Level S-4 with an Acceptable Quality Level (AQL) of 4.0. Sampling is not permitted for the leak tests and the flame tests which shall be performed on each torch offered for acceptance.

4.2.3 Comparison tests. When specified (see 6.2.1), comparison tests shall be performed on one torch drawn at random from each month's production. The comparison tests shall consist of the tests of 4.4.1 and 4.4.4 through 4.4.7. Failure of a torch to pass any comparison test shall be cause for deferring further submission of torches for Government acceptance until the contractor has completed the following actions, and three successive randomly selected torches from the represented month's production have completed the comparison tests without failure. The contractor shall:

(a) Conduct a failure analysis study, performing a dimensional, physical, and visual examination, as required, of the component(s) or condition(s) suspected to be the cause of failure.
(b) Evaluate and correct the applicable production processes and procedures to prevent recurrence of the same defect in future production.
(c) Examine torches awaiting submittal for Government acceptance, partially assembled torches, and components (including components and subassemblies at. in-process or final assembly) to insure that material containing the same defect(s) is purged from the inventory and not presented to the Government for acceptance.

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(d) Provide objective evidence of the results of the failure analysis and corrective actions to the Government representative prior to submitting the reconditioned torches for retest.

4.3 Product examination. Visually and manually examine the torches to determine conformance with the requirements of 3.2, 3.3, 3.3.1 or 3.3.2 (as applicable), 3.3.3 through 3.3.3.6, and 3.5.1 through 3.9. Visual examination shall include verification of completeness of manufacture and assembly, comformance to specified standards, adequacy of markings, proper cleaning, and freedom from defects. Manual examination shall include the operation of movable parts by hand to assure proper functioning. Examination provisions may be applied at the earliest practical point in manufacture at which it is feasible to inspect for acceptance without risk of change in the characteristic by subsequent operations. Failure of the contractor to provide objective evidence that the torches and their component parts have passed the examinations prescribed for them by the contractor's inspection system shall be cause for rejection.

4.4 Tests.

4.4.1 Flow capacity test. The Type I torch with cutting attachment or the Type II torch, as applicable, shall be attached to test equipment consisting of regulated supplies of oxygen and acetylene, each with a flow meter and pressure gage, in series, in its delivery line. The pressure gages shall be located next to the torch inlets between the torch check valves and the flow meters. A cutting tip of the largest size supplied with the torch in accordance with 3.5.5.5 shall be inserted in the cutting head. The oxygen flow capacity of the torch, with the cutting oxygen valve fully open, shall then be measured at the torch oxygen inlet pressure specified in 3.4.1.1. The torch shall then be lit and adjusted for a neutral flame, with the cutting oxygen valve closed and the acetylene regulator adjusted to produce the flow rate specified in 3.4.1.2. The pressure at the torch acetylene inlet shall then be measured. Failure of the torch to maintain the flow rates specified in paragraphs 3.4.1.1 and 3.4.1.2 within the required pressure limits shall be cause for t-ejection. NOTE: The accuracy of gas flow measurement devices is affected by changes in the temperature, pressure, and specific gravity of the flowing gas. Flowmeter readings must be adjusted by the appropriate factors to compensate for any variations between the test conditions and those for which the flowmeter was calibrated.

4.4.2 Leak tests. Each torch snail be tested for leaks using an inlet gas pressure of not less than 50 psig. At the manufacturer's option, any method of leak detection with a sensitivity equal to or greater than that of a water immersion bubble leak test may be used. Any leak in the torch which cannot be stopped either by tightening a valve with a torque of 20 inch-pounds or less, or by tightening a threaded connection or valve packing nut with the wrench provided with the torch, as well as any leak which cannot be stopped without tightening a valve packing nut to the point that the valve cannot be turned with a torque of 20 inch-pounds or less, shall be considered an incurable leak. For first article testing only, the cutting oxygen valve shall be opened and closed by the cutting oxygen lever at least 1000 times prior to the leak tests. The following tests shall be performed.

4.4.2.1 Type T torches. Type I torches shall be tested for leaks at SO psig or more as follows:

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(a) With both inlet valves on the torch handle closed and no obstruction to gas passage at the front body, both inlets shall be pressurized to check the torch handle for valve seat leaks.
(D) with the front body plugged and both inlet valves open, both inlets shall be pressurized to check the torch handle for external leaks.
(c) with the preheat and cutting oxygen valves closed and no obstruction to gas passage at the cutting head, the cutting attachment shall be installed on the torch handle or other equivalent fixture. The attachment oxygen inlet shall then be pressurized to check for valve seat leaks.
(d) With the preheat and cutting oxygen valves open and the cutting head plugged, the attachment oxygen and acetylene inlets shall be pressurized to check the attachment for external leaks.

Any incurable leak, detected during the course of the test, sufficient to produce a bubble in ten seconds in a water immersion test shall be cause for rejection (see 3.4.3 and 3.5.5.3).

4.4.2.2 Type II torches. Type II torches shall be tested for leaks at 50 psig or more as follows:

(a) With all valves closed and no obstruction to gas passage at the cutting head, both inlets shall be pressurized to check for valve seat leaks.
(b) With the cutting head plugged and all valves open, both inlets shall be pressurized to check the torch for external leaks.

Any incurable leak, detected during the course of the test, sufficient to produce a bubble in ten seconds in a water immersion test shall be cause for rejection (see 3.4.3 and 3.5.5.3).

4.4.3 Flame tests.

4.4.3.1 Torches. Each Type I torch, with its cutting attachment installed, and each Type II torch shall be tested as follows for leaks at the cutting head and for flame stability:

(a) With a cutting tip attached, the torch shall be lit and adjusted to produce a neutral flame at the gas pressures recommended for the tip by the manufacturer. Repeated popping or backfire of the flame is indicative of an internal leak and shall be cause for rejection.
(b) The cutting oxygen valve shall then be fully opened, and the preheat flame observed for changes. Other than elongation, any distortion of the flame is indicative of an internal leak and shall be cause for rejection.
(c) A pilot light or other equally sensitive method shall be used to check for leaks at the tip nut. Any leak at the tip nut shall be cause for rejection.
(d) For torch mix torches, the end of the cutting tip shall be tapped against a firebrick, carbon block, or other hard surface to produce a succession of five backfires at a rate of at least one per second. Any flashback, as Indicated by a pale blue flame accompanied by a sustained high-pitched whistle and possibly sparks issuing from the flame holes and/or by the handle becoming hot to the touch, shall be

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cause for rejection (see 3.4.2). WARNING: If a flashback does occur, the oxygen valve must be closed immediately, then the acetylene valve must be closed. If the gas valves are not closed quickly, permanent damage to the torch and/or injury to personnel may result.

4.4.3.2 TipS. Each cutting tip and each welding tip shall be flame tested as follows:

(a) The tip shall be attached to a torch or other equivalent fixture and a flame shall be lit and adjusted to produce a neutral flame at the pressures recommended for the tip by the manufacturer. The flame shall be visually examined. Failure of the tlp to produce a symmetrical flame shall be cause for rejection.
(b) A pilot light or other equally sensitive method shall be used to check for leaks at the tip seat. Any leak at the tip seat shall be cause for rejection.
(c) For all cutting tips, the cutting oxygen stream shall be turned on and the path of the oxygen stream (visible as a dark line through the flame) examined. Failure of the cutting oxygen stream to pass straight through the flame within two degrees, as indicated by comparison with a visual standard, shall be cause for rejection.
(d) The end of each tip mix cutting tip shall be tapped against a firebrick, carbon block, or other hard surface to produce a succession of five backfires at a rate of at least one per second. Any flashback shall be cause for rejection.

4.4.4 Valve lever bend test. The Type I torch with cutting attachment or the Type II torch, as applicable, shall be secured in a vise or other suitable fixture so that the torch is in the horizontal position with the cutting oxygen valve lever on top. A point load of at least 30 pounds shall be applied straight downward on the center of the lever at a point at least four inches from the nearest supporting part (lever fulcrum or valve stem), or within 1/2 inch of the end of the lever if it is less than four inches long. Any permanent bend in the lever caused by the application of the load shall be cause for rejection (see 3.5.5.4).

4.4.5 Valve knob pull test. The Type I torch with cutting attachment or the Type 11 torch, as applicable, shall be secured in a vise or other suitable fixture. A force of at least 70 pounds, directed away from the torch along the centerline of the valve stem, shall be applied to each valve knob in turn. Failure of any knob to withstand the 70 pound pull without loosening or coming off shall be cause for rejection (see 3.5.1.2.4).

4.4.6 Valve knob torque test. With the valve stem held stationary, a clockwise and a counter-clockwise torque of at least 50 inch-pounds shall be applied in each direction to each valve knob in turn. Failure of any knob to withstand the 50 inch-pound torque without loosening or coming off shall be cause for rejection (see 3.5.1.2.4). At the manufacturer’s option, the valve knob and stem assemblies may be removed from the torch for this test to prevent damage to the valve seats.

4.4.7 Gas tube strength test. The Type I cutting attachment or the Type II torch, as applicable, shall be secured in a vise or other suitable fixture so that the gas tubes are supported no closer than five inches from the centerline

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of the tip nut, The tip nut shall be tightened with a torque of no less than 40 foot-pounds applied at 90 25 degrees to the tubes. Failure of the gas tubes to withstand the torque as specified in 3.5.5.1 shall be cause for rejection.

4.5 Packaging inspection. Packaging inspection shall be conducted before and after packaging to determine compliance with the requirements of Section 5.

5. PACKAGING

5.1 Military preservation, packing, and marking. When specified (see 6.2.1), requirements for Level A preservation, Levels A and B packing, and military marking shall be in accordance with MIL-W-45562.

5.2 Industrial preservation, packing, and marking. Unless otherwise specified (see 6.2.1), preservation, packing, and marking shall be in accordance with ASTM D3951-82.

5.3 Thread protection. Gas inlet threads shall be protected from damage during handling, storage, and shipments with plastic caps conforming to MIL-C-5501.

6 NOTES

6.1 Intended use. The apparatus (torch assemblies) covered by this specification is intended for use by the military for field maintenance operations in support of a diverse representation of military equipment, including tactical wheeled and tracked vehicles and weapon systems. The apparatus is intended for use with acetylene fuel gas and may frequently be used by personnel relatively inexperienced at welding and cutting. The equipment will most often be used in unsheltered areas, and may likely be stored in dusty or damp environments.

6.2 Ordering data.

6.2.1 Acquisition requirements. Acquisition documents should specify the following:

(a) Title, number, and date of this specification.
(b) Type of torch required (see 1.2).
(c) First article when required (see 3.1).
(d) Quantity and size of welding tips required if different (see 3.5.4.1).
(e) Quantity and size of cutting tips required if different (see 3.5.5.5).
(f) Exceptions and additional safety/health requirements when required (see 3.7).
(g) product identification if different (see 3.8).
(h) First article inspection if different (see 4.2.1).
(i) Comparison tests when required (see 4.2.3). (NOTE: Comparison testing is not appropriate for contract quantities of less than 500 torches.)
(j) Level of preservation, packing, and marking if different (see 5.1 and 5.2).

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6.3 Contract data requirements. Required technical data such as operator's manuals, parts lists, and other instructions for operation and maintenance, as identified on a numbered DD Form 1664, should be specified on a DD Form 1423 incorporated in the contract.

6.4 Definitions. Terms used in this specification are defined as follows:

(a) Backfire - backfire is defined as the explosion of gases due to a momentary retrogression of the flame into the torch tip, usually caused by snuffing the burning tip against any surface or by an imbalance of gases. A typical backfire (a loud "pop") results in immediate, automatic extinguishing of the flame and, if an ignition source exists outside of the tip, the mixed gases then promptly relight and burn properly.

(b) Flashback - flashback is defined as a retrogression of the flame into the torch as far back as the mixing chamber and a sustained burning of gases inside the torch or tip.

(c) Positive pressure - positive pressure is defined as acetylene pressure from one to fifteen psig.

6.5 Safety and health requirements. In order that equipment integrated into the user's operational environment will comply with OSHA limitations and control of noise levels, radiation, electromagnetic emission, noxious vapors, heat, etc, as applicable, specific requirements concerning such points of operation, and other health and safety requirements, should be specified by the user.

6.6 Changes from previous issue. Asterisks are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

Custodians:
Army - AL
Navy - YD
Air Force - 99

Review activity:
Air Force - 84

User activity:
Navy - MC

Preparing activity:
Army - AL

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-T-13880H	Torches, Oxyacetylene, Cutting and Welding

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR
b. ADDRESS *(Street, City, State, ZIP Code)*	☐ USER
	☐ MANUFACTURER
	☐ OTHER *(Specify)* _____

5. PROBLEM AREAS

a. Paragraph Number and Wording.

b. Recommended Wording

c. Reason/Rationale for Recommendation

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

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